

[illegible]

[illegible]

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	Date	Open	High	Low	Close	Adj Close	Volume
0	2012-01-04	895.0	908.0	870.0	875.0	678.447571	70300
1	2012-01-05	875.0	881.0	863.0	878.0	680.773499	46300
2	2012-01-06	870.0	877.0	855.0	870.0	674.570618	28800
3	2012-01-10	871.0	893.0	871.0	884.0	685.425781	65600
4	2012-01-11	884.0	930.0	879.0	928.0	719.541809	131100

```
df['Volume'].value_counts()
```

Volume

```
0      24
27700  12
38900  12
66200  11
49300  10
```

```
..
281000  1
144900  1
10300   1
8500    1
227000  1
```

Name: count, Length: 1157, dtype: int64

```
cat_col = 'Volume'
```

```
label_encoder = LabelEncoder()
```

```
df['Volume_label_encoded'] = label_encoder.fit_transform(df[cat_col])
```

```
mapping = dict(zip(label_encoder.classes_,
                    label_encoder.transform(label_encoder.classes_)))
```

```
print("Соответствие значений и меток:")
```

```
# print(mapping)
```

```
df[[cat_col, 'Volume_label_encoded']].head()
```

Соответствие значений и меток:

	Volume	Volume_label_encoded
0	70300	587
1	46300	354
2	28800	182
3	65600	542
4	131100	970

Задача 24

```
volume = df['Volume']
```

```
lower_bound = volume.quantile(0.05)
```

```
upper_bound = volume.quantile(0.95)
```

```
print("Нижняя граница по 5% квантилю:", lower_bound)
print("Верхняя граница по 95% квантилю:", upper_bound)

outliers = df[(volume < lower_bound) | (volume > upper_bound)]

print("\nОбнаруженные выбросы:")
print(outliers)

df_filtered = df[(volume >= lower_bound) & (volume <= upper_bound)]

print("\nНабор данных после удаления выбросов:")
print(df_filtered)
```

Нижняя граница по 5% квантилю: 16700.0
Верхняя граница по 95% квантилю: 144424.99999999999

Обнаруженные выбросы:

	Date	Open	High	Low	Close	Adj Close	Volume
\							
7	2012-01-16	926.0	990.0	918.0	990.0	767.614807	217000
8	2012-01-17	1000.0	1027.0	994.0	1017.0	788.549683	184400
9	2012-01-18	1014.0	1014.0	910.0	919.0	712.563599	255600
10	2012-01-19	919.0	928.0	888.0	900.0	697.831421	168800
14	2012-01-25	909.0	955.0	909.0	955.0	740.476929	165900
...	...	...	...	...	...	...	...
3037	2024-05-01	2449.0	2460.0	2405.0	2414.0	2414.000000	10300
3038	2024-05-02	2415.0	2440.0	2406.0	2414.0	2414.000000	8500
3041	2024-05-09	2352.0	2391.0	2350.0	2377.0	2377.000000	14100
3042	2024-05-10	2392.0	2392.0	2361.0	2374.0	2374.000000	9900
3044	2024-05-14	2352.0	2592.0	2333.0	2590.0	2590.000000	227000

	Volume_label_encoded
7	1111
8	1076
9	1129
10	1060
14	1053
...	...
3037	15
3038	8

```
3041          42
3042          13
3044        1117
```

[304 rows x 8 columns]

Набор данных после удаления выбросов:

	Date	Open	High	Low	Close	Adj Close	Volume
\							
0	2012-01-04	895.0	908.0	870.0	875.0	678.447571	70300
1	2012-01-05	875.0	881.0	863.0	878.0	680.773499	46300
2	2012-01-06	870.0	877.0	855.0	870.0	674.570618	28800
3	2012-01-10	871.0	893.0	871.0	884.0	685.425781	65600
4	2012-01-11	884.0	930.0	879.0	928.0	719.541809	131100
...	...	...	...	...	...	...	...
3047	2024-05-17	2500.0	2545.0	2480.0	2545.0	2545.000000	23600
3048	2024-05-20	2545.0	2604.0	2545.0	2572.0	2572.000000	27900
3049	2024-05-21	2572.0	2581.0	2507.0	2514.0	2514.000000	23700
3050	2024-05-22	2524.0	2551.0	2520.0	2539.0	2539.000000	17800
3051	2024-05-23	2552.0	2566.0	2525.0	2558.0	2558.000000	24400

	Volume_label_encoded
0	587
1	354
2	182
3	542
4	970
...	...
3047	130
3048	173
3049	131
3050	73
3051	138

[2748 rows x 8 columns]